Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

STARK

Mechanical Design Engineer

Dear STARK team,

I'd like to apply for the Mechanical Design Engineer role. Taking a part from a sketch to SolidWorks, a 3D printed prototype and then a manufactured, tested assembly is the work I enjoy most, and doing it for autonomous flight systems would be a big and exciting step.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton. A variable stiffness mechanism, with a worm gear that pretensions the spring, works in parallel with a 6 Nm motor. I optimised the mechanism with a genetic algorithm (diameter, spring stiffness and pretension force), made the parametric models, assemblies and drawings in SolidWorks, 3D printed the first parts and had the final ones manufactured. I integrated the motor, encoder and force/torque sensors and verified its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM) and estimated its stiffness with less than 10% error, so working out loads, stiffness and trade-offs is part of my normal work. In my Master's I worked in a team of 4 electronics and mechanical engineers on a soft exosuit, and in my Bachelor's I did the CAD, machining and assembly of an arm exoskeleton and exo glove, and built a Go-Kart and a hybrid Efficycle with student teams. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me sketch concepts and explain trade-offs clearly.

I would love to bring new and innovative ideas to your UAV platforms and grow with your design team. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder